(U) ODEsSA Interface Control Document



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POC:	Mike Lesniak, NGIC
Contact:	usarmy.charlottesville.ngic.list.modsimoffice@mail.mil

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(U) FOR TITLE PAGE
(U) TIMELINE: MOST ITEMS: Ready by SRR, finalized by PDR. REVISION NUMBERS & DATES: Finalized at FDR.
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(U) FOR REVISION TABLE:
(U) TIMELINE: Each milestone submission needs its own row, in 0.# format. Final delivery is 1.0. Dates are YYYYMMDD.

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Revision	Date	Author	Description of Changes
1.0	20240908	BAH	New Template
1.1	20241226	BAH	Updates to template based on NGIC feedback
1.2	20250115	BAH	Updates to template based on NGIC feedback
3.0	20250217	BAH	Baselined to 3.0 release for document templates
4.0	20251114	BAH	Baselined to 4.0 release for document templates

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1 (U) Introduction

(U) TIMELINE: This information should be filled out by PDR.
(U) For each described DLL specify which version of the architecture is associated with it.
(U) If your model is not being wrapped for integration, this document can simply state that the model is only designed to interface with the ODESSA Scenario GUI and currently does not support external integration.

(U) This document describes the use of the compiled DLL of the X.Y.Z version of the Overarching Dynamic Electronic-Warfare System Standard Architecture (ODESSA) framework. Booz Allen Hamilton (BAH) creates models for use with ODESSA in support of the National Ground Intelligence Center (NGIC) to facilitate integration of the model and ODESSA into external simulation environments.

2 (U) Threat System Description

(U) Ideally the ICD is unclassified. The "Reference ..." sentence below directs the reader to the classified UAM instead of describing the system here.

(U) Reference the ModelNameGoesHere Users' and Analysts' Manual, Section #, for a description of the threat system.

3 (U) Model External Simulation Interface

(U) This section describes the functions and corresponding structures of the external simulation interfaces that are specific to the ModelNameGoesHere DLL. For information regarding the behavior of the model itself, please see the ModelNameGoesHere User and Analyst Manual (UAM).

3.1 (U) Model DLL Functions

(U) TIMELINE: The table should be ModelNameGoesHere finalized by CDR. It should include the intended functions accessed through the DLL.

Table 1: (U) DLL Functions

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Function
Example: InitFxn()
GetTxGain
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Continued on next page

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Table 1: (U) DLL Functions (Continued)

UNCLASSIFIED
Function
GetRxGain
...
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3.1.1 (U) EXAMPLE InitModelNameGoesHereSystem

(U) TIMELINE: Each of the model's external frameworks, including all wrapper functions and structs, should be built and documented by Preliminary Design Review (PDR). Final state of those should exist by Critical Design Review (CDR).
(U) The ModelNameGoesHere initialization function. Prepares the ModelNameGoesHere for use.

Table 2: (U) Function call name here

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Input Parameter	Description	Output Parameter	Description
const std::string	Full path to ODESSA scenario file	Void	-
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(U) Create a new section using this format for each function listed in DLL Functions Table.

3.2 (U) Model DLL Structs

(U) TIMELINE: The table should be populated by System Requirements Review (SRR) and finalized by CDR. It should include the intended data structures accessed and used through the DLL.

Table 3: (U) DLL Structs

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Structure
Example: ModelExports()
ModelImports
...
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3.2.1 (U) EXAMPLE ModelNameGoesHereOutputs

(U) TIMELINE: Each of the descriptor sections should exist by PDR. Final state of those should exist by CDR.

(U) The ModelNameGoesHere output data structure.

Table 4: (U) Struct name here

UNCLASSIFIED			
Input Parameter	Description	Output Parameter	Description
void	-	const std::string	Full path to ODESSA scenario file
			UNCLASSIFIED

(U) Create a new section using this format for each function.

4 (U) Sources

(U) Reference the UAM for a comprehensive list of sources used to develop this model. Reference Citations below for those sources used in creating this Interface Control Document (ICD).

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5 (U) Acronyms

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Term	Description	Page
BAH	*Booz Allen Hamilton*	1
CDR	*Critical Design Review*	2, 3
DLL	*dynamic link library*	v, 1, 2
ICD	*Interface Control Document*	3
NGIC	*National Ground Intelligence Center*	1
ODESSA	*Overarching Dynamic Electronic-Warfare System Standard Architecture*	1
PDR	*Preliminary Design Review*	2, 3
SRR	*System Requirements Review*	2
UAM	*User and Analyst Manual*	1, 3

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